



Knowledge Institute of Technology

Centre for Career Development & Training (CDT)

Summative Assessment I (Pen & Paper Mode)

C Programming - Silicon Batch 6 (SI - 6)

Date: 29.06.2026 Duration: 1 Hr. 15 Mins.

Max.Marks: 100 Batch : 2027 / IV Year

Section – A (60 Minutes)

Programming Questions (4 * 20 = 80)

1. In a race, each participant's finish time is recorded. You need to find out who finished first and how long it took them. Additionally, you need to find the average finish time of all participants.

Input Format

A list of floating-point numbers representing finish times.

Enter number of participants:

5

Enter finish times:

12.5

10.0

11.2

13.3

9.8

Output Format

- Two floating-point numbers: the first-place time and the average time.

First Place Time = 9.80

Average Time = 11.36

2. You are given two integers A and B. Your task is to find and return an integer representing the value of their bitwise OR operation.

The bitwise OR operator returns:

- $1 \text{ OR } 0 = 1$
- $0 \text{ OR } 1 = 1$
- $0 \text{ OR } 0 = 0$
- $1 \text{ OR } 1 = 1$

Sample Input

5

3

Sample Output

7

3. You are given a function, `void MaxInArray(int arr[], int length);` The function accepts an integer array 'arr' of size 'length' as its argument. Implement the function to find the maximum element of the array and print the maximum element and its index to the standard output (STDOUT). The maximum element and its index should be printed in separate lines.

Note:

- Array index starts with 0
- Maximum element and its index should be separated by a line in the output
- Assume there is only 1 maximum element in the array
- Print exactly what is asked, do not print any additional greeting messages

Example:

Input:

23 45 82 27 66 12 78 13 71 86

Output:

86

9

Explanation:

86 is the maximum element of the array at index 9.

4. Given an array of N integers and an integer D, rotate the array by D positions in the specified direction.

- If the direction is L, rotate the array to the left.
- If the direction is R, rotate the array to the right.

If D is greater than the size of the array, rotate the array by $D \% N$ positions. Your task is to return the rotated array. This is a classic array manipulation problem frequently used in coding interviews to assess understanding of array indexing and in-place transformations.

Input Format

- The first line contains an integer N, the size of the array.
- The second line contains N space-separated integers.
- The third line contains an integer D, representing the number of positions to rotate.
- The fourth line contains a character L or R, indicating the rotation direction.

Output Format

Print the elements of the rotated array separated by a single space.

Constraints

- $1 \leq N \leq 10^5$
- $0 \leq D \leq 10^9$
- $-10^9 \leq \text{arr}[i] \leq 10^9$

Sample Input 1

```
5
1      2      3      4      5
2
L
```

Sample Output 1

```
3 4 5 1 2
```

Explanation:

Rotating the array two positions to the left moves the first two elements to the end.

Sample Input 2

```
7
10      20      30      40      50      60      70
3
R
```

Sample Output 2

```
50 60 70 10 20 30 40
```

Explanation:

Rotating the array three positions to the right moves the last three elements to the beginning.

(Contd. to Section B)

Section – B (15 Minutes)
Aptitude Questions (10 * 2 = 20)

- 1) A car travels from City P to City Q at a speed of 60 km/hr and returns from Q to P at a speed of 40 km/hr. What is the average speed of the car for the entire journey?
A) 45 km/hr B) 48 km/hr C) 50 km/hr D) 52 km/hr E) 55 km/hr
- 2) A train 240 metres long is running at a speed of 72 km/hr. In how many seconds will it cross a platform 360 metres long?
A) 20 seconds B) 24 seconds C) 28 seconds D) 30 seconds E) 36 seconds
- 3) A boat covers 32 km downstream in 4 hours and the same distance upstream in 8 hours. What is the speed of the boat in still water?
A) 3 km/hr B) 4 km/hr C) 5 km/hr D) 6 km/hr E) 8 km/hr
- 4) A sum of money amounts to ₹9,440 in 3 years and ₹10,400 in 5 years at simple interest. What is the principal sum?
A) ₹7,800 B) ₹7,900 C) ₹8,000 D) ₹8,060 E) ₹8,180
- 5) Find the compound interest on ₹15,000 for 2 years at 10% per annum, compounded annually.
A) ₹3,000 B) ₹3,100 C) ₹3,150 D) ₹3,200 E) ₹3,300
- 6) In an examination, a student scored 35% marks and failed by 12 marks. Another student scored 50% marks and obtained 18 marks more than the minimum passing marks. What is the maximum mark of the examination?
A) 180 B) 190 C) 195 D) 200 E) 210
- 7) A shopkeeper marks an article 40% above its cost price and then allows a discount of 15% on the marked price. What is his overall profit percentage?
A) 16% B) 18% C) 19% D) 21% E) 24%
- 8) Five friends — Manoj, Neha, Om, Priya, and Rahul — are ranked based on their salaries. Neha earns more than Om but less than Priya. Rahul earns more than Priya. Manoj earns less than Om. Who earns the highest salary?
A) Manoj B) Neha C) Om D) Priya E) Rahul
- 9) Six people — J, K, L, M, N, and O — sit in a row facing north. K sits third from the left end. M sits immediately to the right of K. L sits at one of the extreme ends. N sits second from the right end. J sits immediately to the left of O. Who sits at the left extreme end?
A) J B) K C) L D) N E) O
- 10) In a class of 40 students, Aman ranks 14th from the top. Sonia ranks lower than Aman, with exactly 5 students positioned between them. How many students rank below Sonia in the class?
A) 18 B) 19 C) 20 D) 21 E) 22
